

Global Education Analysis: Kristoffer Fields

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Project Overview: This project utilizes Python and R to analyze and examine the relationship between GDP per capita and literacy rate. The data used in this project is from the World Bank database as well as the UCDP database. The objective of this research is to analyze the outliers in the data in this relationship and explore other variables as confounders.

Variable indicators:

Net enrollment in primary education

Adult Literacy Rate

GDP Per Capita

Conflict

Time range: 1970 - 2020

Introduction

Comparing these countries also allows us to see how educational progress connects with broader economic indicators like GDP. Through this, we gain a clearer understanding of how various parts of the world are progressing in terms of educational development. GDP per capita is widely used as a standard indicator of economic development because it provides a simple measure of a country's average income level and reflects its overall productive capacity. Higher GDP per capita typically correlates with greater resources available for public services, including education, healthcare, and infrastructure, which is why it is often used to compare living standards across countries. Oftentimes, GDP per capita can be misleading because it does not effectively capture how evenly income is distributed effectively. For example, a country may have a high GDP but poor literacy due to income inequality. Due to this, researchers often opt for using the Human Development Index.

Why is this important?

Education is one of the most integral and fundamental parts of life. Access to quality education from a young age is key to human development and future opportunity. By keeping track of global educational trends, we can provide insight into how countries are performing and also discover where support or policy reform is necessary. Beyond this, studying education alongside economic indicators such as GDP per capita helps us understand how national wealth translates into human development. Through analyzing these relationships, policymakers can evaluate where economic growth is not improving educational outcomes and where targeted interventions may be needed.

Data Visualization and Analyses

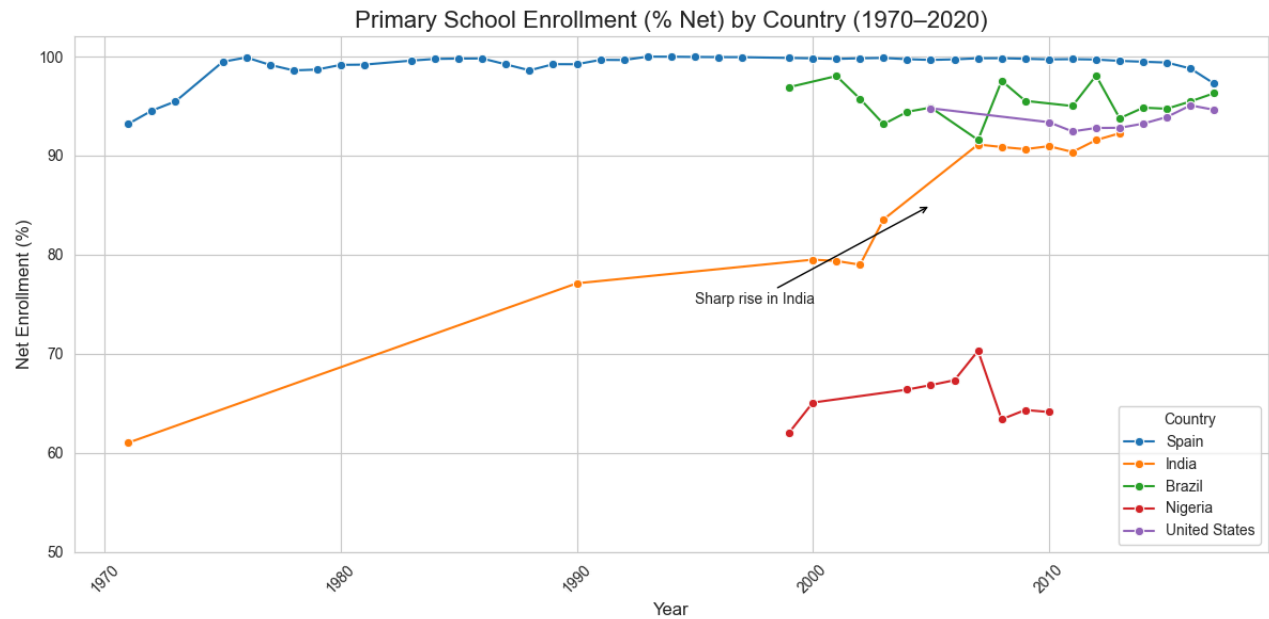


Figure 1

This line graph displays the net primary school enrollment by country from 1970 to 2020. The graph also displays several interesting trends. For example, in 2000, the primary school enrollment percentage in India substantially increased, with a sharp rise. It is important to note that enrollment in general has been increasing in India over the past 50 years; however, the rate of increase has noticeably accelerated since 2000. At the same time, other developing countries like Nigeria, continue to lag behind despite similar levels of economic classification, signaling persistent barriers to educational progress. This is an interesting comparison to countries like Spain and The United States who have been over 90 percent for quite some time.

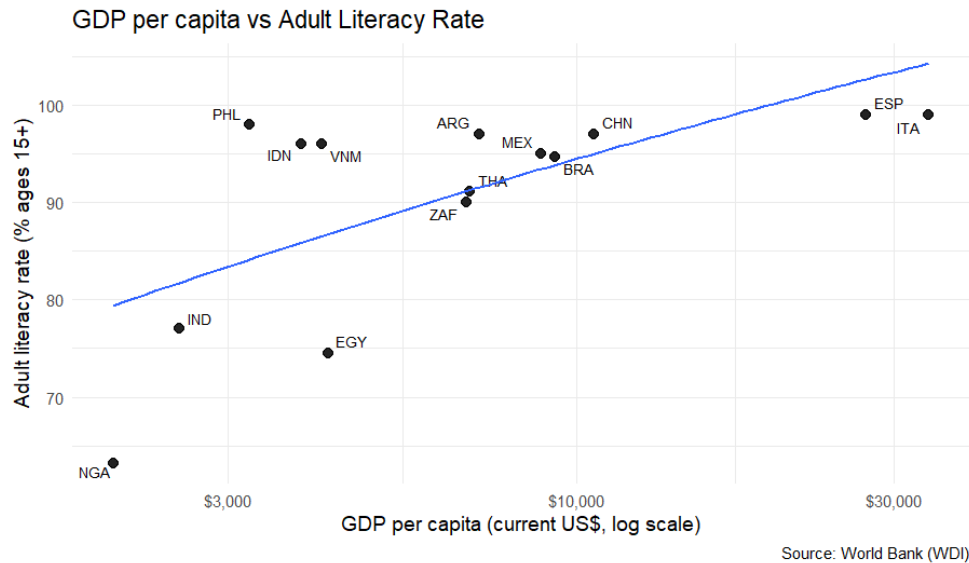
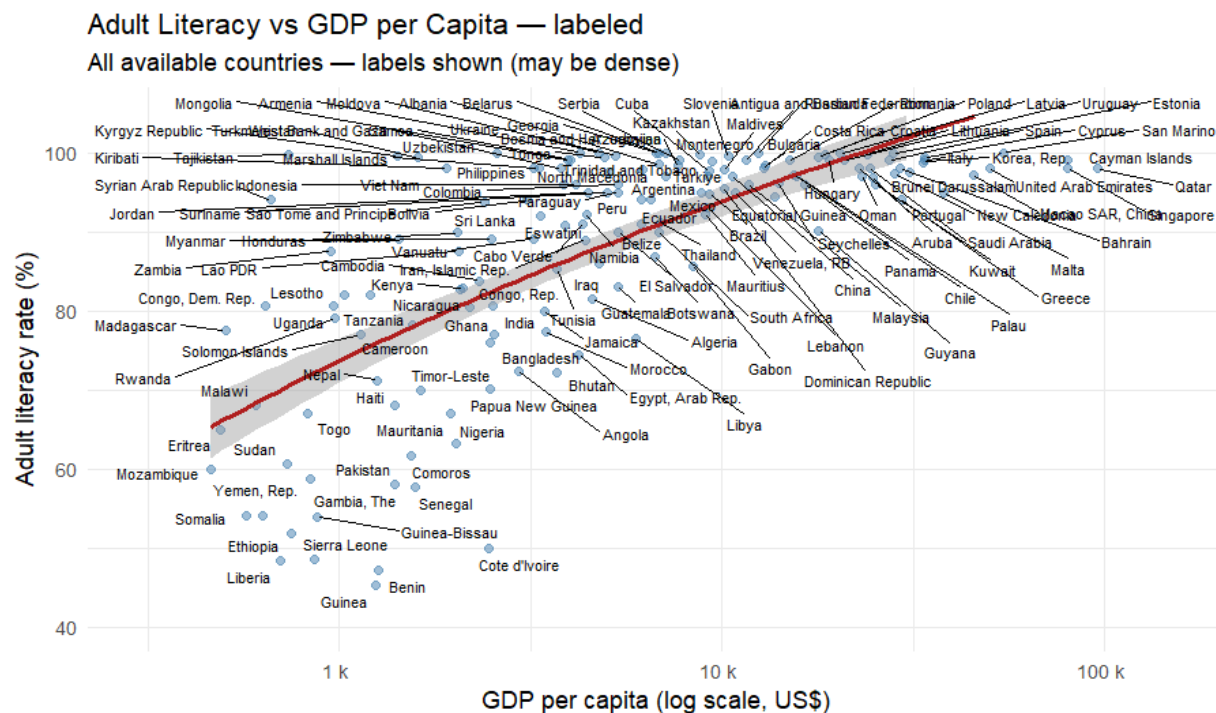


Figure 2

Figure 2 shifts from enrollment trends to the broader relationship between national wealth and educational outcomes by comparing GDP per capita with adult literacy rates. The positively slope regression indicates a strong relationship between wealth and literacy rate. Additionally, the scatter displays meaningful variation. Countries such as Spain and Italy cluster in the top-right, reflecting both high income and near-universal literacy, this is clearly supported by the relationship presented between GDP and literacy. These overperformers lie above or near the line, this is a reflection of adequate investment in resources and human capital. This connection is strongly associated with the comparison between Spain and the United States and why they have been successful in their own regard. However, countries like Nigeria and Egypt underperform, demonstrating how political instability and a lack of investment by the government can negatively affect outcomes. If we add conflict to the figure further it will show under-performing countries are also those experiencing minor or major conflict, highlighting conflict as a key confounding factor in literacy outcomes. This connects to the comparison between Bulgaria and El Salvador.

Figure 3



The regression analysis determines a strong correlation between GDP per capita and literacy rate according to World Bank data. The linear model created by expressing GDP per capita on a logarithmic scale helps us understand the distribution of education and variation in literacy, with an R squared of 0.48. Furthermore, the coefficient of approximately 9.76 indicates that a one-unit increase in GDP per capita is roughly equivalent to a proportional increase in income. Additionally this is associated with nearly a ten percentage-point rise in literacy. This finding further proves the point that wealthier nations with more resources and infrastructure achieve higher literacy outcomes. In terms of the scatter plot, the upward sloping curve reinforces this research, while there are some outliers, it is important to understand why this is the overall trend of the data.

Figure 4

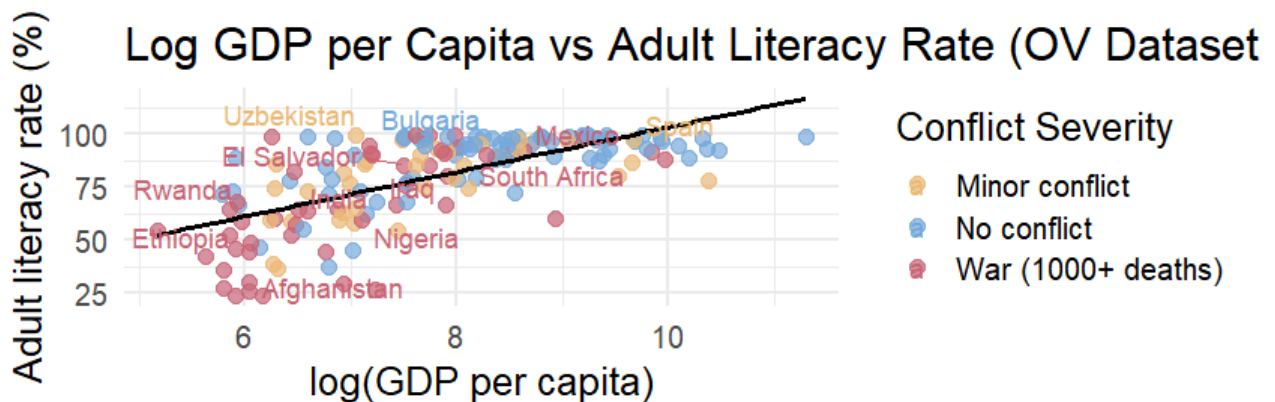


Figure 4 visualizes the strong positive correlation between economic prosperity and literacy outcomes. It uses a log-linear model based on World Bank indicators from 1970 to 2020. The logarithmic transformation of GDP per capita enables a more realistic depiction of cross-country variation by compressing the substantial income differences between developing and industrialized economies. The resulting regression line shows a clear upward trajectory: countries with higher average incomes tend to have substantially higher literacy rates.

Case 1: Spain and The United States

The United States and Spain represent two high-income democracies within the developed world, yet they differ significantly in terms of economic scale, demographic dynamics, and historical context. Both exhibit near-universal literacy and political stability, but the composition of their economies and their approaches to social welfare diverge substantially. While the United States functions as a continental-scale economy with vast domestic production and a global reserve currency, Spain operates within the European Union, constrained by the eurozone's fiscal framework yet benefiting from integration with a large common market. These structural distinctions shape how each country translates wealth into educational access, social

equity, and overall human development. It is essential to understand how these conditions relate to the number of educated people, despite the vastly different cultural conditions and ways of life.

Both countries boast adult literacy rates near 99%, but their educational cultures differ. Spain’s literacy expansion was rapid after Franco, driven by massive public investment and EU cohesion policies that emphasized inclusive education. According to the UCDP data, neither country faces active internal conflict. While the most recent conflict in Spanish history would be the Basque movement in 2011, the U.S. experiences high rates of firearm violence and political polarization, but not an organized conflict exceeding the UCDP threshold. In essence, the scatter plot and data show the correlation between a strong economic presence and educational attainment, and peace.

Indicator	United States us	Spain es	Source / Year
GDP per capita (current US\$)	\$80,400	\$35,300	World Bank (2023)
GDP per capita, PPP (Intl \$)	\$80,700	\$56,200	IMF (2023)
Adult literacy rate (%)	99%	99%	UNESCO (2022)
Human Development Index (HDI)	0.93	0.91	UNDP (2023)
Life expectancy (years)	79.3	83.2	WHO (2023)
Gini coefficient (inequality)	0.41	0.33	World Bank (2021)
Unemployment rate (%)	3.6%	11.7%	ILO (2023)
Education expenditure (% GDP)	5.0%	4.3%	UNESCO (2022)
Public debt (% GDP)	121%	110%	IMF (2023)
Conflict status (UCDP ±2 y)	No conflict	No conflict	UCDP/PRIO ACD (2024)

Table 1

Case 2: Uzbekistan and South Africa

South Africa and Uzbekistan offer an interesting comparison because both maintain strong primary school enrollment, but for very different reasons. South Africa's enrollment has remained relatively high over the past several decades. However, the country still deals with the long-term effects of apartheid, which created deep inequalities in access and educational quality. Since the 1990s, enrollment has become more stable as the government expanded schooling for the entire population, but literacy levels remain uneven, especially when comparing different social and economic groups.

Uzbekistan, on the other hand, inherited a highly structured education system from the Soviet Union, which emphasized universal schooling and literacy. As a result, Uzbekistan has historically maintained extremely high enrollment and literacy rates, even through economic transitions after independence in 1991. Although Uzbekistan's GDP per capita is lower than South Africa's, its education outcomes are generally more consistent because basic services are more evenly distributed. South Africa's higher income does not automatically translate into better educational results due to inequality, while Uzbekistan's strong institutional legacy allows it to perform above what its economic level might predict. Together, these countries highlight how historical and political structures can shape educational success just as much as a nation's wealth. South Africa and Uzbekistan offer an interesting comparison, South Africa's enrollment has generally stayed in the 92–97% range since the early 2000s, although the quality of schooling differs sharply across regions. Literacy rates hover around 85–90%, which is solid but still reflects deep inequality between urban and rural communities. Uzbekistan, meanwhile, has consistently posted enrollment rates above 95%, and its literacy rate, shaped by the strict Soviet education system, has remained around 99% for decades, making it one of the highest in the

world. Even though South Africa's GDP per capita is roughly 2–3 times higher, Uzbekistan's education outcomes appear more uniform because access to basic schooling is more evenly distributed. This contrast shows that wealth alone doesn't guarantee stronger educational performance; the structure of institutions and historical investment patterns matter just as much. In other words, South Africa is wealthier, but it is unevenly distributed.

Case 3: Bulgaria and El Salvador

Bulgaria and El Salvador are two countries that show very different paths in education because of their history and economic situations. I chose these two countries because they clearly illustrate how countries with comparable income levels can produce very different educational results depending on their institutional history, political stability, and exposure to conflict. In Figure 3, Bulgaria performs slightly above our predicted value and El Salvador strongly underperforms for its income, despite both countries having a history of conflict. The key difference between this comparison and the prior one about South Africa and Uzbekistan is the fact that the systemic institutional legacies and powers are what shaped these countries. On the other hand, El Salvador and Bulgaria have deeper political issues such as instability and conflict which define their relationship with GDP per capita and literacy.

Bulgaria, which is part of the European Union, has had strong government involvement in schools for a long time, so its primary enrollment rates stay high, usually around 94 to 97 percent. The country also has a very high adult literacy rate of about 98 percent. With a GDP per capita between 12,000 and 14,000 dollars, Bulgaria has been able to invest in updated classrooms, better teacher training, and more modern learning programs. El Salvador has made progress, too, but it deals with more challenges, including inequality, migration, and political

instability. Its primary enrollment rates are still solid, generally between 88 and 93 percent, but adult literacy is lower at around 88 to 90 percent. Since the GDP per capita is only around 4,000 dollars, schools in El Salvador often have fewer resources, and the quality of education can vary a lot depending on the region.

Conflict does not directly sit on the y-axis in this graph, but it still matters as a confounding factor because countries with chronic violence tend to underfund schools, lose teachers, and experience disrupted attendance. Bulgaria's long-term political stability after the early two thousand removes that drag on its educational system, while El Salvador has had to deal with violence that shapes how much it can invest in human capital. So even though both countries have improved economically over time, their historical environments help explain why Bulgaria achieves much higher literacy outcomes than El Salvador at similar or even lower levels of income. The comparison shows how economic growth alone is not enough; stability, institutional capacity, and the absence of conflict play major roles in shaping long-term educational results.